

CHEMICAL CHANGES – STRONG & WEAK ACIDS

KEY VOCABULARY

Strong acid: fully ionised in water, e.g. HCl, H₂SO₄.

Weak acid: only partially ionised, e.g. ethanoic, citric.

Concentration: the amount of acid dissolved per dm³.

Ionisation: the splitting of a molecule into ions in water.

COMMON MISCONCEPTIONS

~~Strong and concentrated mean the same.~~

✓ Strong = fully ionised; concentrated = lots dissolved.

~~Weak acids cannot be concentrated.~~

✓ A weak acid can still be very concentrated.

~~Weak acids do not fully react.~~

✓ They react fully, just more slowly / lower H⁺.

STRONG VS WEAK

Strong: HCl → H⁺ + Cl⁻ (fully ionised)

Weak: CH₃COOH ⇌ CH₃COO⁻ + H⁺ (partial)

Same concentration → strong acid has lower pH.

pH falls by 1 = H⁺ concentration ×10

Strength ≠ concentration.

1 DEFINE STRONG

State what is meant by a **strong acid**. [1 mark]

2 STRONG VS CONCENTRATED

Explain the difference between a strong acid and a concentrated acid. [2 marks]

3 COMPARE PH

Two acids have the same concentration. **Explain** why the strong acid has a lower pH. [2 marks]

4 PH AND H⁺

If pH falls from 4 to 2, state how the H⁺ concentration changes. [1 mark]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

A strong acid is ____ ionised; a weak acid is only ____ ionised.

fully

partially

strong

weak

6 IDENTIFY

Classify each as strong or weak: HCl, citric acid, H₂SO₄. [3 marks]